MACHINE LEARNING LAB

III B. TECH- II SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A5AI20	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES

1. Make use of Data sets in implementing the machine learning algorithms
2. Implement the machine learning concepts and algorithms in any suitable language of choice.
3. To impart knowledge on the basic concepts underlying machine learning.
4. To acquaint with the process of selecting features for model construction.
5. To familiarize different types of machine learning techniques.

COURSE OUTCOMES

Upon successful completion of the course, the student is able to

1. Design and implement machine learning solutions to classification, regression problems.
2. Analyze the complexity of Machine Learning algorithms and their limitations.
3. Apply appropriate data sets to the Machine Learning algorithms.
4. Identify and apply Machine Learning algorithms to solve real world problems
5. Apply supervised and unsupervised techniques on various data sets.

WEEK – I BASICS

Write a program to demonstrate the following
a) Operation of data types in Python.
b) Different Arithmetic Operations on numbers in Python.
c) Create, concatenate and print a string and access substring from a given string.
d) Append, and remove lists in python.
e) Demonstrate working with tuples in python.
f) Demonstrate working with dictionaries in python.

WEEK- II STATISTICAL OPERATIONS

Using python write a NumPy program to compute the
a) Expected Value
b) Mean
c) Standard deviation
d) Variance
e) Covariance
f) Covariance Matrix of two given arrays.

WEEK- III DATA PREPROCESSING – CONTINUOUS / DISCRETE DATA

For a given set of training data examples stored in a .CSV file, demonstrate Data Preprocessing in Machine learning with the following steps
a) Getting the dataset.
b) Importing libraries.
c) Importing datasets.
d) Finding Missing Data.
e) Finding Outliers
f) Splitting dataset into training and test set.
g) Feature scaling.

WEEK- IV DATA PREPROCESSING – CATEGORICAL DATA

For a given set of training data examples stored in a .CSV file, demonstrate Data Preprocessing in Machine learning with the following steps
a) Getting the dataset.

b) Importing libraries.
c) Importing datasets.
d) Finding Missing Data.
e) Encoding Categorical Data.
f) Splitting dataset into training and test set.
g) Feature scaling.

WEEK- V **DECISION TREE**

Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample

WEEK- VI **LINEAR REGRESSION**

Build a linear regression model using python for a particular data set by
a) Splitting Training data and Test data.
b) Evaluate the model (intercept and slope).
c) Visualize the training set and testing set
d) predicting the test set result
e) compare actual output values with predicted values

WEEK- VII **MULTIPLE LINEAR REGRESSION**

Build a multiple linear regression model using python for a particular data set by
a) Splitting Training data and Test data.
b) Evaluate the model (intercept and slope).
c) Visualize the training set and testing set
d) predicting the test set result
e) compare actual output values with predicted values

WEEK- VIII **LOGISTIC REGRESSION**

The dataset contains information of users from a company's database. It contains information about UserID, Gender, Age, EstimatedSalary, and Purchased. Use this dataset for predicting that a user will purchase the company's newly launched product or not by Logistic Regression model.

User ID	Gender	Age	EstimatedSalary	Purchased
15624510	Male	19	19000	0
15810944	Male	35	20000	0
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	0
15728773	Male	27	58000	0
15598044	Female	27	84000	0
15694829	Female	32	150000	1
15600575	Male	25	33000	0
15727311	Female	35	65000	0
15570769	Female	26	80000	0
15606274	Female	26	52000	0
15746139	Male	20	86000	0
15704987	Male	32	18000	0
15628972	Male	18	82000	0
15697686	Male	29	80000	0
15733883	Male	47	25000	1
15617482	Male	45	26000	1
15704583	Male	46	28000	1

WEEK- IX **CLUSTERING**

A python program to implement K-Means, Hierarchical Clustering and PCA.

WEEK- X **KNN & SVM**

Write a Python program to implement KNN and SVM

WEEK- XI **NAIVE BAYES**

Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.

WEEK- XII **RANDOM FOREST**

Implement Random Forest Algorithm using Python.

TEXT BOOKS

1. Aurélien Géron - Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd Edition. September 21019, O'Reilly Media, Inc., ISBN: 9781492032649. 2. Tom Mitchel "Machine Learning", Tata McGraW Hill, 2017.